DISCO-TAB revision — table review

This file shows the three revised tables in the JBHI/IEEE two-column geometry. Table I (datasets), Table II (aggregate benchmark), Table III (DISCO-TAB vs. best competitor).

TABLE I
BENCHMARK DATASETS (23 TOTAL: 9 FROM THE ORIGINAL STUDY PLUS 10 CLINICAL TIME-TO-EVENT COHORTS FROM SURVSET AND 4 FROM TABARENA). TIME-TO-EVENT COHORTS USE THE EVENT INDICATOR AS A BINARY TARGET.

Dataset	**Domain**	**#Inst.**	**#Feat.**	**#Cls.**	**Source**
Heart Failure	Clinical	299	12	2	[**?**]
Breast Cancer	Clinical	569	30	2	[**?**]
Liver Disorders	Clinical	345	5	2	[**?**]
Parkinsons	Clinical	195	22	2	[**?**]
Obesity	Clinical	2111	16	7	[**?**]
Diabetes	Clinical	100000	8	2	[**?**]
German Credit	Finance	1000	20	2	[**?**]
Bank Marketing	Finance	45211	16	2	[**?**]
Census Income	Finance	48842	14	2	[**?**]
Melanoma	Clinical	205	5	2	[**?**]
PBC	Clinical	312	6	2	[**?**]
PBC3	Clinical	349	12	2	[**?**]
Retinopathy	Clinical	368	7	2	[**?**]
WHAS500	Clinical	461	16	2	[**?**]
GBSG2	Clinical	686	8	2	[**?**]
Colon	Clinical	885	9	2	[**?**]
GRACE	Clinical	998	5	2	[**?**]
TRACE	Clinical	1878	6	2	[**?**]
Framingham	Clinical	4699	7	2	[**?**]
Blood Transfusion	Clinical	533	4	2	[**?**]
QSAR Biodeg	Chemistry	1052	41	2	[**?**]
Maternal Risk	Clinical	452	6	3	[**?**]
Website Phishing	Security	724	9	3	[**?**]

TABLE II

BENCHMARK SUMMARY: MEAN OVER ALL DATASETS WITH 95% CI (SAME CELL, SMALLER FONT) FOR EACH METRIC AND GENERATOR, GROUPED BY MODEL FAMILY. **BOLD** = BEST PER ROW AMONG THE FULL-COVERAGE MODELS. ARROWS GIVE THE PREFERRED DIRECTION. *Ranges:* ALL SCORES LIE IN $[\mathbf{0}, \mathbf{1}]$ EXCEPT DCR, MMD AND WASSERSTEIN, WHICH ARE UNBOUNDED ($\geq \mathbf{0}$, SCALE DEPENDS ON THE DATASET), AND PRDC DENSITY, WHICH MAY EXCEED $\mathbf{1}$; MIA AUROC AND C2ST ARE AUROCS FOR WHICH $\mathbf{0.5}$ IS IDEAL (SYNTHETIC INDISTINGUISHABLE FROM REAL). THE #DATASETS ROW GIVES PER-MODEL COVERAGE; '–' MARKS A METRIC NOT EVALUATED FOR THAT MODEL.

		Ours	**GAN**				**VAE**		**Diff.**	**Stat./Flow**			**LLM**		
	Metric	**DISCO-TAB**	**CTGAN**	**ADS-GAN**	**PATE-GAN**	**TAEGAN**	**TVAE**	**RTVAE**	**TabDDPM**	**GaussCop**	**NFlow**	**Marg.**	**GReaT**	**RTF**	**TAGAL**
	#datasets	23	23	23	23	23	23	23	23	23	23	23	12	1	5
Utility (TSTR)	F1-macro ↑	**0.637** ±.06	0.470 ±.06	0.620 ±.05	0.423 ±.04	0.638 ±.05	0.403 ±.13	0.578 ±.05	0.603 ±.07	0.543 ±.06	0.538 ±.06	0.409 ±.05	0.509 ±.05	0.795	0.614 ±.14
	F1-weight. ↑	0.679 ±.06	0.549 ±.08	**0.687** ±.05	0.530 ±.07	0.709 ±.05	0.453 ±.14	0.665 ±.05	0.683 ±.07	0.640 ±.06	0.632 ±.07	0.456 ±.06	0.632 ±.09	0.827	0.690 ±.14
	Accuracy ↑	0.685 ±.06	0.578 ±.08	0.706 ±.05	0.583 ±.07	0.728 ±.05	0.485 ±.14	0.688 ±.06	**0.714** ±.06	0.670 ±.07	0.666 ±.06	0.464 ±.06	0.663 ±.10	0.833	0.670 ±.13
Fidelity	JS div. ↓	0.043 ±.02	0.088 ±.03	0.070 ±.02	0.174 ±.05	0.059 ±.03	0.132 ±.03	0.232 ±.07	0.084 ±.06	0.047 ±.03	**0.035** ±.01	0.208 ±.04	0.040 ±.01	0.013	0.071 ±.05
	KS stat. ↓	**0.078** ±.01	0.217 ±.04	0.183 ±.03	0.349 ±.06	0.147 ±.04	0.267 ±.04	0.248 ±.04	0.120 ±.06	0.108 ±.03	0.112 ±.02	0.389 ±.06	0.109 ±.03	0.052	0.162 ±.08
	Helling. ↓	0.138 ±.02	0.197 ±.05	0.184 ±.04	0.298 ±.05	0.165 ±.05	0.300 ±.04	0.347 ±.08	0.172 ±.07	**0.125** ±.04	0.136 ±.02	0.356 ±.05	0.140 ±.03	0.087	0.201 ±.09
	Corr. ↑	0.919 ±.02	0.821 ±.06	0.913 ±.02	0.797 ±.05	0.892 ±.03	0.844 ±.03	0.865 ±.04	0.886 ±.03	**0.929** ±.02	0.862 ±.04	0.818 ±.05	0.913 ±.03	0.939	0.839 ±.11
Privacy	DCR ↑	0.340 ±.08	0.556 ±.13	0.357 ±.09	0.612 ±.16	0.310 ±.08	0.255 ±.07	0.334 ±.06	0.763 ±.52	0.399 ±.09	0.404 ±.10	**1.420** ±.40	0.262 ±.12	0.684	0.449 ±.29
	NNDR ↑	0.886 ±.04	0.872 ±.04	0.871 ±.04	0.882 ±.04	0.878 ±.03	0.949 ±.02	**0.953** ±.02	0.814 ±.07	0.846 ±.05	0.857 ±.05	0.868 ±.05	0.839 ±.09	0.880	0.925 ±.03
	MIA →.5	0.525 ±.01	0.503 ±.01	0.512 ±.01	**0.499** ±.00	0.514 ±.01	0.502 ±.01	0.512 ±.01	0.537 ±.03	0.501 ±.01	0.503 ±.01	0.501 ±.01	0.504 ±.02	0.444	0.490 ±.05
	Dup. ↓	0.017 ±.04	**0.003** ±.00	0.020 ±.03	0.008 ±.01	0.034 ±.05	0.029 ±.04	0.019 ±.03	0.062 ±.09	0.006 ±.01	0.016 ±.03	0.003 ±.00	0.000 ±.00	0.000	0.040 ±.03
Diversity	Precis. ↑	0.866 ±.08	0.615 ±.12	0.818 ±.10	0.511 ±.12	0.874 ±.09	**0.896** ±.09	0.820 ±.09	0.808 ±.12	0.824 ±.09	0.728 ±.13	0.441 ±.12	0.956 ±.02	0.987	0.983 ±.02
	Recall ↑	0.782 ±.09	0.895 ±.04	0.857 ±.03	0.839 ±.08	0.849 ±.05	0.323 ±.09	0.611 ±.11	**0.967** ±.02	0.929 ±.03	0.941 ±.02	0.918 ±.08	0.815 ±.10	0.954	0.801 ±.09
	Density ↑	0.929 ±.18	0.414 ±.11	0.806 ±.20	0.341 ±.15	1.040 ±.24	**1.134** ±.22	0.950 ±.22	0.930 ±.26	0.682 ±.12	0.681 ±.20	0.227 ±.10	1.033 ±.08	1.152	1.187 ±.29
	Coverage ↑	0.769 ±.11	0.525 ±.11	0.710 ±.10	0.365 ±.10	0.807 ±.10	0.395 ±.08	0.625 ±.11	**0.876** ±.11	0.774 ±.11	0.703 ±.13	0.223 ±.11	0.795 ±.11	0.987	0.626 ±.38
Detection	C2ST ↓	0.819 ±.05	0.934 ±.03	0.909 ±.05	0.966 ±.02	0.898 ±.05	0.960 ±.02	0.978 ±.03	**0.726** ±.09	0.815 ±.07	0.873 ±.06	0.972 ±.02	0.744 ±.11	0.442	0.842 ±.13
	MMD ↓	0.023 ±.01	0.100 ±.03	0.040 ±.01	0.238 ±.08	0.018 ±.01	0.176 ±.04	0.048 ±.02	0.074 ±.07	0.021 ±.02	**0.019** ±.00	0.393 ±.09	0.032 ±.02	0.000	0.041 ±.05
	Wass. ↓	**0.128** ±.02	0.414 ±.09	0.263 ±.04	0.664 ±.12	0.175 ±.04	0.444 ±.06	0.314 ±.05	0.583 ±.41	0.173 ±.07	0.178 ±.01	1.755 ±.68	0.206 ±.05	0.094	0.275 ±.18
SDMetrics	Quality ↑	0.659 ±.08	0.561 ±.06	0.631 ±.08	0.495 ±.06	0.658 ±.08	0.548 ±.07	0.554 ±.07	0.662 ±.10	**0.666** ±.08	0.624 ±.07	0.462 ±.06	0.619 ±.13	0.438	0.506 ±.18
	Shapes ↑	0.823 ±.03	0.729 ±.04	0.764 ±.04	0.626 ±.05	0.795 ±.04	0.674 ±.04	0.715 ±.05	0.784 ±.05	**0.828** ±.02	0.810 ±.03	0.584 ±.05	0.791 ±.04	0.875	0.789 ±.07
	Pairs ↑	0.495 ±.16	0.394 ±.12	0.498 ±.16	0.364 ±.11	0.520 ±.15	0.423 ±.14	0.393 ±.13	**0.539** ±.19	0.504 ±.15	0.438 ±.13	0.339 ±.11	0.448 ±.25	0.000	0.223 ±.34

TABLE III

DISCO-TAB VS. THE BEST ESTABLISHED COMPETING GENERATOR PER METRIC (MEAN OVER ALL DATASETS, WITH 95% CI). **BOLD** = WINNER. SCORES ARE IN $[\mathbf{0}, \mathbf{1}]$ EXCEPT DCR/MMD/WASSERSTEIN (UNBOUNDED $\geq \mathbf{0}$) AND PRDC DENSITY ($> \mathbf{1}$ POSSIBLE); MIA AND C2ST ARE AUROCS WITH $\mathbf{0.5}$ IDEAL.

	Metric	DISCO-TAB	Best comp.	Method
Utility (TSTR)	F1-macro ↑	**0.637** ±.06	0.620 ±.05	ADS-GAN
	F1-weight. ↑	0.679 ±.06	**0.687** ±.05	**ADS-GAN**
	Accuracy ↑	0.685 ±.06	**0.714** ±.06	**TabDDPM**
Fidelity	JS div. ↓	0.043 ±.02	**0.035** ±.01	**NFlow**
	KS stat. ↓	**0.078** ±.01	0.108 ±.03	GaussCop
	Helling. ↓	0.138 ±.02	**0.125** ±.04	**GaussCop**
	Corr. ↑	0.919 ±.02	**0.929** ±.02	**GaussCop**
Privacy	DCR ↑	0.340 ±.08	**1.420** ±.40	**Marg.**
	NNDR ↑	0.886 ±.04	**0.953** ±.02	**RTVAE**
	MIA →.5	0.525 ±.01	**0.499** ±.00	**PATE-GAN**
	Dup. ↓	0.017 ±.04	**0.003** ±.00	**CTGAN**
Diversity	Precis. ↑	0.866 ±.08	**0.896** ±.09	**TVAE**
	Recall ↑	0.782 ±.09	**0.967** ±.02	**TabDDPM**
	Density ↑	0.929 ±.18	**1.134** ±.22	**TVAE**
	Coverage ↑	0.769 ±.11	**0.876** ±.11	**TabDDPM**
Detection	C2ST ↓	0.819 ±.05	**0.726** ±.09	**TabDDPM**
	MMD ↓	0.023 ±.01	**0.019** ±.00	**NFlow**
	Wass. ↓	**0.128** ±.02	0.173 ±.07	GaussCop
SDMetrics	Quality ↑	0.659 ±.08	**0.666** ±.08	**GaussCop**
	Shapes ↑	0.823 ±.03	**0.828** ±.02	**GaussCop**
	Pairs ↑	0.495 ±.16	**0.539** ±.19	**TabDDPM**